UGC NET ENVIRONMENTAL SCIENCE · PAPER 2

UNIT 3: ENVIRONMENTAL BIOLOGY

Complete Study Notes — Part 1 of 2 (Topics 1–5)

Ecology · Autecology · Population Ecology · Community Ecology · Succession

Highest priority unit | ~12-14 Qs/exam | Part 1 of 2 | Score 100%

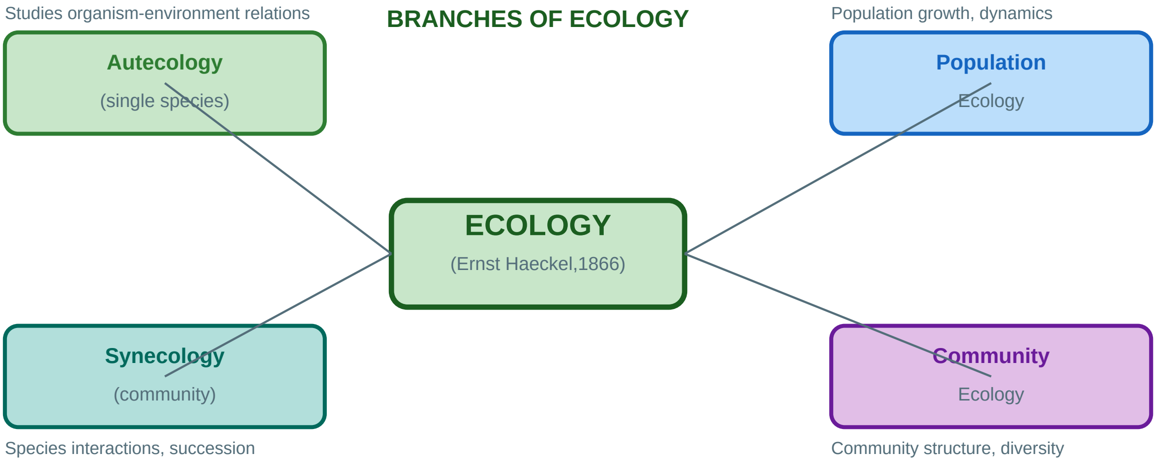
EXAM OVERVIEW — UNIT 3 AT A GLANCE

Parameter	Detail
Unit name	Unit 3 — Environmental Biology
Priority	HIGHEST among all 10 units — single largest source of questions
Questions in exam	12–14 questions per paper (out of 100) = 24–28 marks
Study time	7–8 days (first priority after Unit 1 foundation)
This document	Part 1 covers Topics 1–5: Ecology basics, Autecology, Population Ecology, Community Ecology, Succession
Part 2 covers	Topics 6–10: Biodiversity, Conservation, Genetics, Evolution, Applied Ecology
Most tested topics	Population growth models, Species interactions, Succession types, Biodiversity indices, IUCN categories

Topic (Part 1)	Sub-topics	Exam weight
1. Scope & definitions	Ecology, autecology, synecology, levels of organisation	Medium
2. Autecology	Ecological tolerance, Liebig's law, Shelford's law, limiting factors	Medium
3. Population ecology	Growth models, age structure, survivorship, life tables, r & K selection	HIGHEST
4. Community ecology	Species interactions, coevolution, competitive exclusion, niche partitioning	HIGH
5. Ecological succession	Primary, secondary, seres, climax, models of succession	HIGH

TOPIC 1: SCOPE, DEFINITIONS & LEVELS OF ORGANISATION

Ecology (coined by Ernst Haeckel, 1866) = the scientific study of the relationships between organisms and their environment. The word comes from Greek: *oikos* (house/habitat) + *logos* (study). It is the study of 'who lives where, why, and with what consequences.'



1.1 Key Definitions — Must Know All

Term	Definition	Coined by/Year	Exam weight
Ecology	Study of interactions between organisms and their biotic & abiotic environment	Ernst Haeckel, 1866	VERY HIGH
Autecology	Study of ecology of a single species or individual organism in relation to its environment	—	HIGH
Synecology	Study of ecology of groups of organisms (communities/populations) forming a functional unit	—	HIGH
Population	Group of individuals of the same species occupying a defined area at the same time and interbreeding	—	VERY HIGH
Community	All populations of different species living and interacting in the same area at the same time	—	HIGH
Ecosystem	Functional unit: community + its abiotic environment, with energy flow and nutrient cycling	A.G. Tansley, 1935	VERY HIGH
Biome	Large-scale terrestrial ecosystem defined by climate (temperature + precipitation) and dominant vegetation	—	MEDIUM
Biosphere	Entire zone of life on Earth; the global sum of all ecosystems	Eduard Suess, 1875	MEDIUM
Habitat	Physical location/address where an organism lives	—	HIGH
Ecological niche	Role/function of an organism in its ecosystem — what it eats, how it interacts, when active	G.E. Hutchinson, 1957	VERY HIGH

1.2 Levels of Ecological Organisation (Hierarchical)

Gene → Individual → Population → Community → Ecosystem → Biome → Biosphere Each level has **emergent properties** not seen at lower levels. Ecology mainly studies population level and above.

Level	Definition	Example	Study focus
Individual (Organism)	Single living unit; smallest ecological unit	One tiger	Autecology: physiology, behaviour, tolerances
Population	Same species, same place, same time	All tigers in Sundarbans	Growth, density, age structure, mortality
Community	All species in one area	All organisms in Sundarbans forest	Interactions, diversity, succession
Ecosystem	Community + abiotic environment	Sundarbans mangrove ecosystem	Energy flow, nutrient cycling
Biome	Regional ecosystem defined by climate	Tropical rainforest biome	Global distribution, climate-vegetation links
Biosphere	Global sum of all ecosystems	Earth's zone of life	Global cycles, climate, evolution

TOPIC 2: AUTECOLOGY — ORGANISM-ENVIRONMENT RELATIONSHIPS

2.1 Ecological Tolerance & the Law of Tolerance (Shelford, 1913)

Shelford's Law of Tolerance (1913): Each organism has a minimum, optimum, and maximum level of tolerance for each environmental factor (temperature, pH, salinity, light, etc.). Survival and reproduction are best at the optimum; they fail below the minimum and above the maximum.

Concept	Definition	Example
Tolerance range	Range of conditions within which organism can survive	Trout: 5–25°C water temperature
Optimum	Condition at which organism performs best	Trout growth best at 15°C
Zone of stress	Near limits; organism survives but stressed	Trout at 4°C or 24°C — survives, stressed
Lethal zone	Beyond tolerance limit; organism cannot survive	Trout below 0°C or above 30°C
Stenoecious	Narrow tolerance for multiple factors	Coral reefs — very narrow T, salinity, pH range
Euryoecious	Wide tolerance for multiple factors	Cockroach, rats, crows
Stenotherm	Narrow temperature tolerance	Polar bear, coral
Eurytherm	Wide temperature tolerance	Humans, cockroaches
Stenohaline	Narrow salinity tolerance	Freshwater trout cannot survive in seawater
Euryhaline	Wide salinity tolerance	Salmon (live in freshwater AND seawater)

2.2 Liebig's Law of Minimum (1840)

Liebig's Law of Minimum (Justus von Liebig, 1840): Plant growth is controlled not by the total resources available, but by the ONE essential resource that is most scarce relative to the organism's needs — the **limiting factor**. Like a barrel — water leaks from the shortest stave, not the total wood.

Aspect	Detail
Principle	Growth is limited by the scarcest essential nutrient, not by total nutrient quantity
Original application	Plant nutrition in agriculture (crop yields)
Broader application	Now applied to any essential resource (light, water, nitrogen, phosphorus)
Limiting nutrient in freshwater	Phosphorus (P) is often the limiting nutrient in freshwater bodies
Limiting nutrient in marine systems	Nitrogen (N) is often limiting in marine environments
Practical use	Identifying the limiting factor allows targeted fertilisation/management
Barrel analogy	Barrel can only hold water to level of shortest stave — whichever nutrient is shortest limits growth

2.3 Key Ecological Factors (Abiotic)

Factor	Organism classification	Key ecological terms
Temperature	Stenothermal vs Eurythermal; Poikilotherm vs Homeotherm; Ectotherm vs Endotherm	Q10 = rate doubles per 10°C rise; Bergmann's rule: body size ↑ in cold climates
Light	Photophilous (sun-loving) vs Sciophilous (shade-tolerant)	Photoperiodism: day length controls flowering, migration, reproduction; PAR: 400–700nm
Water/Humidity	Xerophytes (dry), Mesophytes (moderate), Hydrophytes (wet), Hygrophytes (humid)	CAM & C4 photosynthesis in xerophytes; stomatal closure; leaf modifications
Soil (Edaphic)	Calcicoles (alkaline soil lovers) vs Calcifuges (acid soil lovers)	pH 6–7 optimal for most crops; CEC = cation exchange capacity; soil texture
Salinity	Stenohaline vs Euryhaline; freshwater (<0.5 ppt) vs marine (~35 ppt) vs brackish	Osmoregulation mechanisms; halophytes tolerate salt (mangroves)
Wind	Anemophilous (wind-pollinated) vs entomophilous (insect-pollinated)	Krummholz = stunted trees at alpine treeline shaped by wind; anemochory = wind dispersal

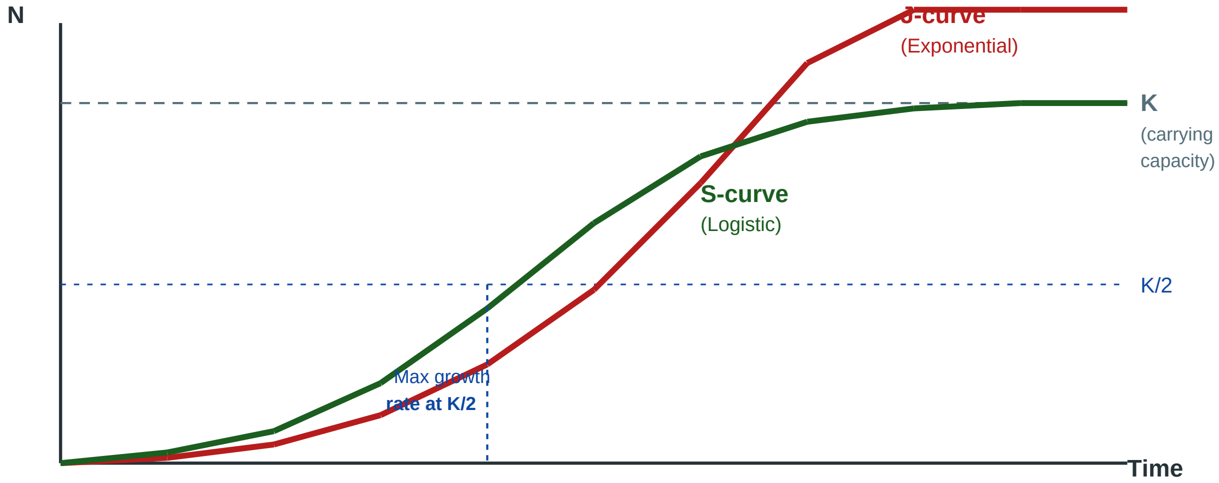
EXAM TIP — Liebig vs Shelford (Frequently Confused)

LIEBIG (1840) = Law of MINIMUM = growth limited by SCARCEST resource. Think: 'Liebig = Least'. SHELFORD (1913) = Law of TOLERANCE = organism has MIN, OPT, MAX for EACH factor. Think: 'Shelford = Spectrum'. Both laws together explain why organisms are not found everywhere — they need each factor within tolerance range, AND none in short supply.

TOPIC 3: POPULATION ECOLOGY — HIGHEST EXAM WEIGHT

3.1 Population Growth Models

POPULATION GROWTH MODELS — J-curve vs S-curve



Parameter	Exponential growth	Logistic growth
Equation	dN/dt = rN	dN/dt = rN(K–N)/K
Curve shape	J-shaped (accelerating)	S-shaped / sigmoidal
Assumes	Unlimited resources, no K	Limited resources; K = carrying capacity
Growth rate	Increases without limit	Increases then decreases; = 0 at N = K
Maximum growth	At maximum N	At N = K/2 (MOST IMPORTANT FACT)
Real world	Early invasion, bacteria in ideal lab	Natural populations with resource limits
r value	High positive r	r can be any value; K limits final size

CRITICAL — Most Repeated Exam Question in Population Ecology

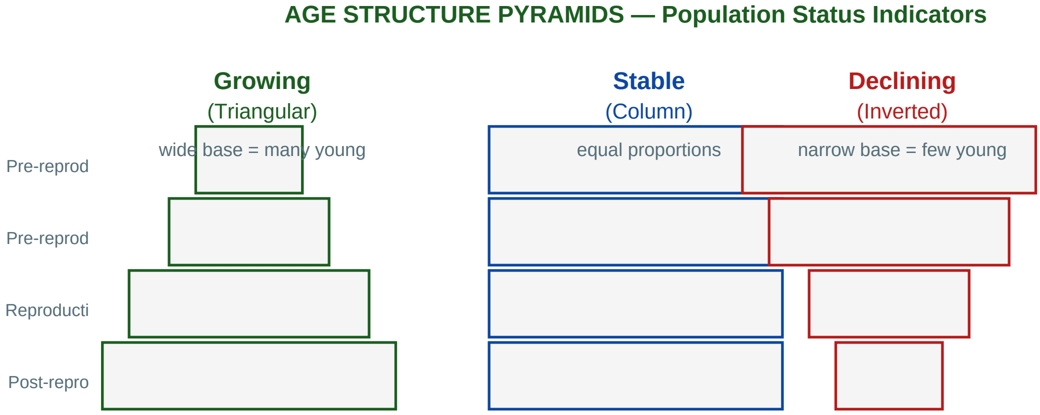
In logistic growth, the MAXIMUM GROWTH RATE occurs when N = K/2. This is also the basis for MAXIMUM SUSTAINABLE YIELD (MSY) in fisheries: fish at N = K/2 for highest sustainable harvest. At N = K → growth rate = 0 (population stable). At N = 0 → growth rate = 0 (no individuals). MAXIMUM is at K/2.

3.2 Key Population Parameters

Parameter	Symbol	Definition	Formula/Notes
Intrinsic rate of increase	r	Max potential growth rate under ideal conditions (biotic potential)	r = birth rate (b) – death rate (d) in ideal conditions
Carrying capacity	K	Max population size environment can sustain	Population stabilises at K in logistic growth
Net reproductive rate	R■	Average female offspring per female per generation	R■>1 growing; R■=1 stable; R■<1 declining
Generation time	T	Average time between mother's birth and daughter's birth	T = Σ(lxmx) / R■

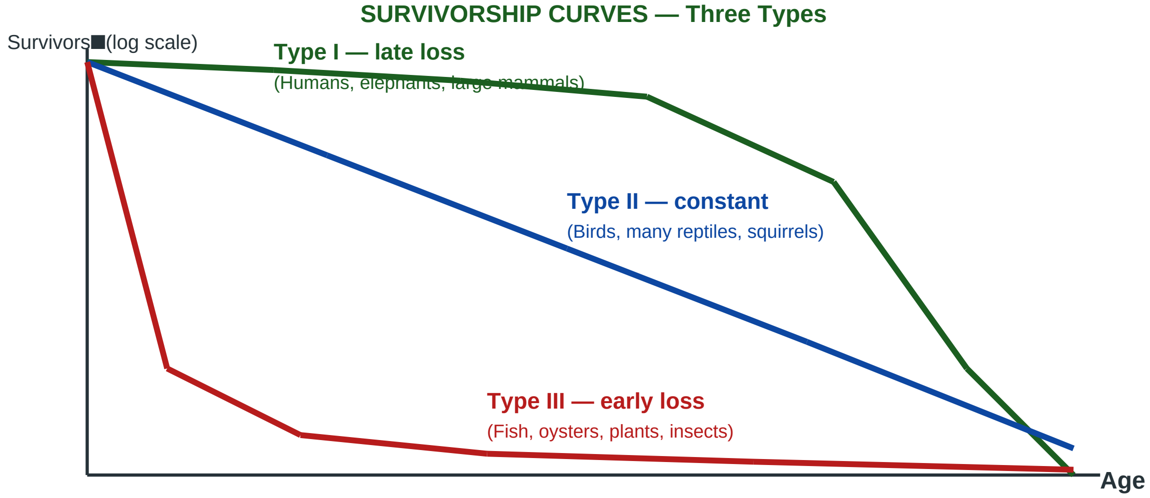
Parameter	Symbol	Definition	Formula/Notes
Finite rate of increase	λ	Population multiplication rate per unit time	λ = e^r; λ>1 growing; λ=1 stable; λ<1 declining
Biotic potential	r_max	Maximum possible r under ideal conditions	Species-specific; elephant r_max very low; bacteria very high

3.3 Age Structure & Population Pyramids



Age pyramid shape	Population status	Pre-reproductive age class	Examples
Triangular (broad base)	Rapidly GROWING	Very large — many young	India (historically), many developing countries, rabbits
Column (rectangular)	STABLE — zero growth	= Reproductive class size	Developed nations (Sweden, Japan historically)
Inverted (narrow base)	DECLINING	Very small — few young	Germany, Italy, Japan (current), some wildlife populations

3.4 Survivorship Curves



Type	Pattern	Mortality timing	Examples
Type I	Convex — survives most of life; steep late mortality	Most die in OLD AGE	Humans, elephants, large mammals, oak trees
Type II	Diagonal — constant mortality at all ages	CONSTANT rate throughout life	Birds (many), squirrels, some reptiles, Hydra

Type	Pattern	Mortality timing	Examples
Type III	Concave — massive early mortality; survivors live long	Most die YOUNG; survivors live long	Fish, oysters, frogs, many insects, annual plants, sea turtles

3.5 r-selection vs K-selection

Character	r-selected (r-strategists)	K-selected (K-strategists)
Body size	Small	Large
Lifespan	Short (often <1 year)	Long (years to decades)
Reproductive rate	HIGH — many small offspring	LOW — few large offspring
Parental care	None or minimal	Extensive, high investment per offspring
Maturation time	Rapid	Slow
Carrying capacity	Far below K	Near K
Survivorship curve	Type III (high early mortality)	Type I (low early mortality)
Habitat preference	Disturbed, unpredictable environments	Stable, predictable environments
Competitive ability	Poor (colonisers)	High (residents)
Examples	Bacteria, insects, annual plants, small rodents, dandelions	Elephants, whales, humans, oak trees, eagles, sharks

3.6 Density-Dependent vs Density-Independent Factors

Type	Definition	Effect with density change	Examples
Density-DEPENDENT (biotic)	Effect INTENSIFIES as population density increases	Stronger effect at high density	Competition, predation, disease, parasitism, intraspecific competition for food/territory
Density-INDEPENDENT (abiotic)	Effect same regardless of population density	Same % mortality regardless of density	Temperature extremes, floods, fires, droughts, hurricanes, volcanic eruptions

Important Population Ecology Concepts

ALLEE EFFECT: at very LOW densities, individual fitness DECREASES — too few mates, reduced cooperative defence, reduced foraging efficiency. Can cause population collapse below critical threshold. MINIMUM VIABLE POPULATION (MVP): smallest population size with >99% survival probability over 1,000 years despite demographic/genetic stochasticity. METAPOPULATION: network of spatially separated subpopulations connected by dispersal. Subpopulations can go extinct locally and be recolonised. Critical for fragmented habitats.

3.7 Life Tables — Key Terms

Symbol	Name	Definition
x	Age class	Age interval (e.g. 0–1, 1–2, 2–3 years)
lx	Survivorship	Probability of surviving from birth to age x
dx	Age-specific mortality	Proportion dying in each age interval
mx	Fecundity	Average number of female offspring per female in age class x

Symbol	Name	Definition
lx × mx	Reproductive contribution	Contribution of age class x to R■
R■	Net reproductive rate	Σ(lx × mx) = average female offspring per female per lifetime

TOPIC 4: COMMUNITY ECOLOGY — SPECIES INTERACTIONS & NICHE

4.1 Species Interactions — Complete Matrix

SPECIES INTERACTIONS — Summary Matrix

Interaction	Effect A	Effect B	Key example
Mutualism	+	+	Mycorrhizae, Rhizobium+legumes, pol
Commensalism	+	0	Epiphytes on tree, barnacles on wha
Amensalism	0	–	Allelopathy: black walnut → juglone
Predation	+	–	Lion–zebra, hawk–mouse, spider–fly
Parasitism	+	–	Tapeworm, Plasmodium, cuckoo (brood
Competition	–	–	Lion vs hyena (interspecific), deer

4.2 Mutualism — Detailed (+/+)

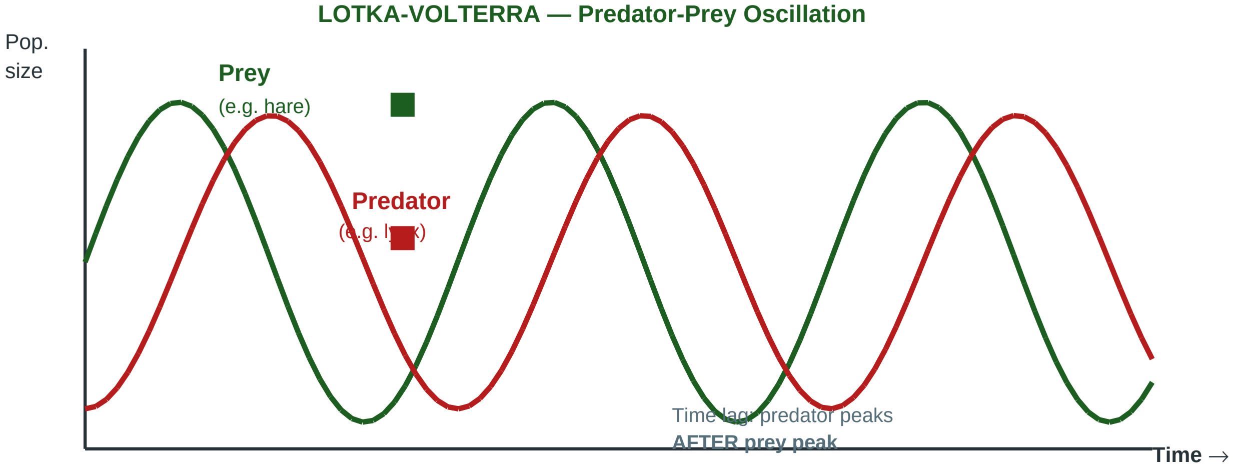
Mutualistic association	Partners	Benefit to each partner	Obligate?
Mycorrhizae	Fungi + plant roots	Fungi get carbohydrates; plant gets water and minerals (especially P)	Often obligate for both
Rhizobium + legumes	N-fixing bacteria + legume plants	Bacteria get shelter + carbohydrates; plant gets fixed nitrogen	Obligate symbiosis
Lichen	Fungus (mycobiont) + algae/cyanobacteria (photobiont)	Fungus gets food; algae gets protection and water retention	Obligate for both
Pollination	Flowering plants + pollinators (bees, butterflies, birds)	Plant gets pollination; pollinator gets nectar/pollen	Variable
Cleaner fish	Cleaner wrasse + client fish	Cleaner gets food (parasites); client gets parasite removal	Facultative
Clownfish + anemone	Clownfish + sea anemone	Clownfish protected from predators; anemone gets nutrients and parasite removal	Facultative
Fig + fig wasp	Fig tree + Agaonid wasp	Wasps lay eggs in fig; fig gets pollinated by wasps	Obligate coevolution

4.3 Predation — Key Concepts (+/–)

Concept	Definition	Example
Predator	Organism that kills and eats other organisms (prey)	Lion, hawk, spider, Venus flytrap
Prey defence	Cryptic colouration (camouflage), aposematism (warning colouration), mimicry, chemical defence	Stick insect, poison dart frog, monarch butterfly
Batesian mimicry	Harmless species mimics harmful species — predator avoids both	Viceroy butterfly mimics toxic monarch butterfly

Concept	Definition	Example
Müllerian mimicry	Two or more unpalatable species mutually mimic each other — reinforces avoidance	Multiple toxic butterfly species with similar orange-black patterns
Aposematism	Bright warning colouration advertising toxicity or danger	Poison dart frogs (bright red/blue/yellow), wasps (yellow-black stripes)
Lotka-Volterra model	Mathematical model of predator-prey oscillation with time lag	Lynx-hare cycle in Canada; classic example of population oscillations

LOTKA-VOLTERRA — Predator-Prey Oscillation



Lotka-Volterra Predator-Prey Equations: Prey: dN/dt = rN – αNP (r = prey growth rate; α = predation rate) Predator: dP/dt = βNP – qP (β = conversion efficiency; q = predator death rate) Key result: populations oscillate cyclically with PREDATOR PEAK LAGGING BEHIND PREY PEAK.

4.4 Competition — Intraspecific vs Interspecific

Type	Between	Outcome	Examples
Intraspecific	Same species	Natural selection; strongest reproduce; can regulate population	Male deer fighting, salmon competing for spawning sites, plant self-thinning
Interspecific	Different species	Competitive exclusion OR niche partitioning OR coexistence	Lions vs hyenas, warblers in spruce trees, Darwin's finches

4.5 Competitive Exclusion Principle & Niche Partitioning

Competitive Exclusion Principle (Gause's Law, 1934): Two species competing for EXACTLY the same ecological niche CANNOT coexist indefinitely in the same habitat. One will outcompete and eliminate the other, OR one will evolve to use a slightly different niche. Demonstrated with two Paramecium species (P. aurelia outcompeted P. caudatum when sharing food).

Concept	Definition	Example
Competitive exclusion	Two species cannot coexist on same niche — one excludes the other	P. aurelia vs P. caudatum in same flask
Niche partitioning (resource partitioning)	Competing species reduce overlap by using different portions of resource	Darwin's finches: different beak sizes → different seed sizes

Concept	Definition	Example
Character displacement	Evolutionary divergence of traits where species coexist to reduce competition	Beak size divergence of Galapagos finches on islands where both coexist
Fundamental niche	Theoretical niche without competition — all conditions species could use	Barnacles can live in all intertidal zones in absence of competition
Realized niche	Actual niche occupied in presence of competitors	Barnacles restricted to upper intertidal zone when outcompeted below

4.6 Parasitism — Key Types (+/–)

Type	Definition	Examples
Ectoparasite	Lives on outside of host	Ticks, lice, fleas, mistletoe on trees
Endoparasite	Lives inside host	Tapeworm, Plasmodium (malaria), Ascaris (roundworm)
Brood parasitism	Parasite lays eggs in host's nest; host raises parasite's young	Common cuckoo (Cuculus canorus) in reed warbler, meadow pipit nests
Hyperparasite	Parasite of a parasite	Viruses parasitising bacteria (bacteriophages); wasps parasitising other wasps
Parasitoid	Insect that lays eggs in/on host; larva kills host during development	Ichneumon wasps, tachinid flies — important biocontrol agents

4.7 Coevolution

Coevolution = reciprocal evolutionary change in two or more interacting species, where each species acts as a selective force on the other. Tight coevolution leads to extreme specialisation. Arms races, mutualistic coevolution, and host-parasite coevolution are major types.

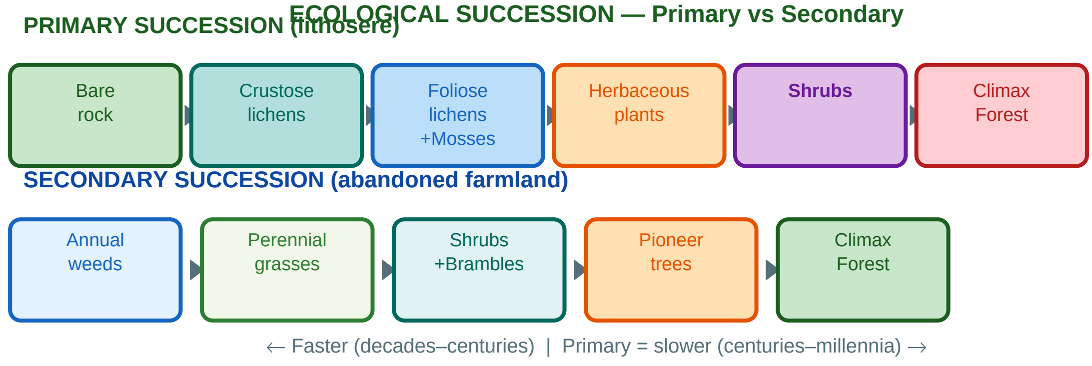
Coevolution type	Description	Classic example
Antagonistic (arms race)	Predator and prey or parasite and host evolve counter-adaptations to each other	Cheetah speed vs gazelle speed; cuckoo egg mimicry vs host rejection behaviour
Mutualistic	Two species evolve together for mutual benefit — increasing specialisation	Fig trees and fig wasps (obligate coevolution); orchids and specific pollinators
Plant-herbivore	Plants evolve defences; herbivores evolve counter-defences	Milkweed toxins vs monarch butterfly resistance; plant thorns vs giraffe tongue

EXAM TIPS — Community Ecology

1. Gause used PARAMECIUM (not plants, not fish) to demonstrate competitive exclusion. 2. Batesian mimicry = HARMLESS mimics HARMFUL. Müllerian = BOTH unpalatable mimic each other. 3. Lotka-Volterra: predator peak LAGS BEHIND prey peak — time delay due to reproductive response. 4. Parasitoids (e.g. Ichneumon wasps) are distinct from parasites — they always kill the host eventually. 5. Obligate mutualists CANNOT SURVIVE without their partner (lichens, fig wasps). Facultative = optional.

TOPIC 5: ECOLOGICAL SUCCESSION — COMPLETE COVERAGE

Ecological Succession = directional, predictable, sequential change in species composition of a community over time at a given location, driven by species modifying their own environment. Ends at climax community in equilibrium with local climate.



5.1 Primary vs Secondary Succession — Complete Comparison

Feature	Primary Succession	Secondary Succession
Starting condition	Bare, lifeless substrate — NO soil, NO organisms	Disturbed area WITH soil and some surviving organisms
Pioneer species	Crustose lichens (most common); also bacteria, algae on bare rock	Annual weeds, grasses, fast-growing herbs
Soil formation	Must develop from scratch via rock weathering (very slow)	Soil already present — just recovering
Duration	Hundreds to thousands of years	Decades to centuries (much faster)
Examples	New volcanic island (Hawaii); exposed glacier; new river delta; abandoned mine	Abandoned farmland; post-fire forest; logged forest; drained wetland
Biological legacy	None initially	Seeds in soil, root systems, soil microorganisms — accelerate recovery
Final climax	Regional climatic climax (same as secondary)	Same regional climax as primary (if undisturbed)
Human relevance	Mine land restoration; wetland creation	Reforestation; conservation of recovering ecosystems

5.2 Types of Succession by Starting Substrate

Type	Substrate	Pioneer organism	Seral stages	Example location
Lithosere	Bare rock	Crustose lichens	Lichen→Moss→Grass→Shrub→Forest	Rocky mountain slopes, quarry faces
Hydrosere / Hydrarch	Open water (lake/pond)	Phytoplankton	Phytoplankton→Submerged plants→Floating plants→Reed swamp→Marsh→Shrub→Forest	Filling lake, pond succession
Psammosere	Sand dunes	Marram grass (Ammophila)	Bare sand→Marram grass→Other grasses→Shrubs→Forest	Coastal sand dunes

Type	Substrate	Pioneer organism	Seral stages	Example location
Halosere	Salt marsh	Glasswort (Salicornia)	Algae→Salicornia→Spartina→Salt marsh→Meadow→Forest	Salt marshes, estuaries
Xerosere	Dry land (general)	Drought-resistant plants	Depends on substrate — includes lithosere, psammosere	Desert margins, dry hillsides

5.3 Seral Stages (Seres)

Seral term	Definition	Key characteristic
Sere	Complete sequence of communities from pioneer to climax	Each sere is characteristic of a habitat type (hydrosere, lithosere etc.)
Seral community (seral stage)	Any intermediate community during succession	Temporary; prepares environment for next stage
Pioneer community	First organisms to colonise bare substrate	Extreme stress-tolerators; modify environment for others
Subclimax community	Community that persists for long time before climax due to local factors	Fire-maintained grasslands, heavily grazed areas
Climax community	Final stable, self-perpetuating community in equilibrium with local climate	Determined by regional climate; high biomass, high complexity
Disclimax (Disturbance climax)	Stable community maintained by recurring disturbance rather than climate	Grassland maintained by fire; heathland maintained by grazing

5.4 Trends During Succession

Property	Change during succession	Explanation
Species diversity	INCREASES	More niches available as habitat complexity increases
Biomass	INCREASES	More organic matter accumulated as succession proceeds
Gross Primary Productivity (GPP)	Increases then stabilises	Maximum in mature forest; levels off at climax
Net Primary Productivity (NPP)	Increases then DECREASES toward climax	More respiration at climax (P:R ratio → 1)
P:R ratio (Photosynthesis:Respiration)	Moves from >1 toward 1 at climax	Climax: GPP ≈ total community respiration
Food web complexity	INCREASES	More species = more feeding links = higher stability
Nutrient cycling efficiency	INCREASES	Less nutrient leaching; more internal recycling
Stability	INCREASES	More species, more interactions = more resilience
Stratification (layers)	INCREASES	Canopy→sub-canopy→shrub→herb→ground layers develop

5.5 Models of Succession (Connell & Slatyer, 1977)

Model	Mechanism	Description	Example
Facilitation	Early → Late species (+)	Early species MODIFY environment making it MORE suitable for later species; early species eventually replaced	Classic Clements model; lichen weathering rock for mosses
Tolerance	Early ≈ Late species (0)	Later species can TOLERATE modified conditions; early species decline as resources consumed	Many plant successions; shade-tolerant climax trees establish under pioneers
Inhibition	Early → Late species (–)	Early species INHIBIT colonisation by later species; succession occurs only as early species die	Algal succession; early colonisers resist replacement

5.6 Retrogressive (Regressive) Succession

Retrogressive succession = backward succession from climax toward earlier seral stages, caused by human activities or natural disturbances. Examples: overgrazing causing forest → grassland → bare soil; deforestation causing reversal; pollution causing ecosystem degradation. Opposite of progressive succession.

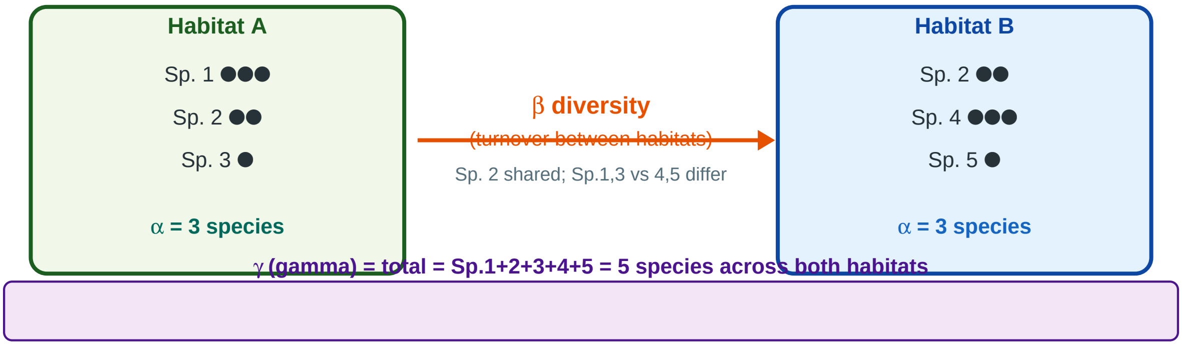
EXAM TIPS — Succession

1. PRIMARY succession = BARE ROCK/substrate. No soil. Pioneer = LICHENS (specifically crustose lichens on rock). Very SLOW. 2. SECONDARY succession = soil present. Pioneer = annual weeds/grasses. FASTER than primary. 3. Hydrarch succession sequence: PHYTOPLANKTON → Submerged plants → Floating plants (Nymphaea) → Reed swamp (Phragmites) → Marsh → Shrubs → FOREST. 4. Psammosere pioneer = Marram grass (Ammophila) — stabilises shifting sand with rhizomes. 5. At CLIMAX: NPP ≈ 0 net (all GPP used in respiration). P:R ratio = 1. Very stable and self-maintaining. 6. NPP DECREASES toward climax (despite GPP being high) — more energy used in respiration by large biomass.

TOPIC 6 PREVIEW: BIODIVERSITY CONCEPTS (Full coverage in Part 2)

6.1 Alpha, Beta, Gamma Diversity — Visual Overview

ALPHA, BETA, GAMMA DIVERSITY — Visual Explanation



Diversity level	Coined by	Definition	Scale	Measurement methods
Alpha (α) diversity	R.H. Whittaker (1960)	Species richness AND evenness within a single community/habitat	Local — one site	Species richness (S), Shannon-Wiener index (H'), Simpson's index (D)
Beta (β) diversity	R.H. Whittaker (1960)	Rate of species TURNOVER/CHANGE between different habitats or communities	Between sites	Whittaker β = γ/α; Jaccard similarity; Sørensen index; Bray-Curtis dissimilarity
Gamma (γ) diversity	R.H. Whittaker (1960)	Total species diversity across a landscape/region (ALL habitats combined)	Landscape/regional	Total species count; γ = α × β (Whittaker relationship)

6.2 Diversity Indices — Formulae

Index	Formula	What it measures	Range/interpretation
Species Richness (S)	S = total number of species	Simply the count of species	Higher S = more diverse; no evenness component
Shannon-Wiener (H')	H' = –Σ(pi × ln pi) where pi = proportion of species i	BOTH richness AND evenness	H' = 0 (one species); higher H' = more diverse; max H' = ln(S)
Simpson's Index (D)	D = Σ(ni(ni–1)) / N(N–1) or 1–D	Dominance; probability two random individuals same species	D near 0 = high diversity; D near 1 = low diversity (one species dominates)
Evenness (Pielou's J)	J = H' / H'max = H' / ln(S)	How evenly individuals distributed among species	J = 0 (uneven) to 1 (perfectly even)

Shannon-Wiener Index — Exam Worked Example

Community with 3 species: Sp.A = 50 individuals, Sp.B = 30, Sp.C = 20. Total N = 100. p_A = 50/100 = 0.5 → p_A × ln(0.5) = 0.5 × (–0.693) = –0.347 p_B = 30/100 = 0.3 → 0.3 × ln(0.3) = 0.3 × (–1.204) = –0.361 p_C = 20/100 = 0.2 → 0.2 × ln(0.2) = 0.2 × (–1.609) = –0.322 H' = –(–0.347 + –0.361 + –0.322) = –(–1.030) = 1.030 Max H' = ln(3) = 1.099. Evenness J = 1.030/1.099 = 0.937 (high evenness).

PART 1 MASTER QUICK REFERENCE — KEY FACTS & MNEMONICS

Scientists & Contributions — Part 1 Topics

Scientist	Year	Contribution	Exam frequency
Ernst Haeckel	1866	Coined 'Ecology' (from Greek oikos+logos)	VERY HIGH
Justus von Liebig	1840	Law of Minimum — growth limited by scarcest nutrient	HIGH
Victor Shelford	1913	Law of Tolerance — min, optimum, max for each factor	HIGH
A.G. Tansley	1935	Coined 'Ecosystem'	VERY HIGH
G.E. Hutchinson	1957	Niche concept: fundamental vs realized; n-dimensional hypervolume	VERY HIGH
G.F. Gause	1934	Competitive exclusion principle; Paramecium experiments	VERY HIGH
Raymond Lindeman	1942	10% law of energy transfer	VERY HIGH
R.H. MacArthur & E.O. Wilson	1967	Island Biogeography Theory	HIGH
R.H. Whittaker	1960	Alpha, beta, gamma diversity; coined terms	HIGH
H.T. Odum	1957	Energy circuit ecology; ecosystem energy flow	MEDIUM
P.H. Leslie	1945	Life table analysis; Leslie matrix for population projection	MEDIUM
Connell & Slatyer	1977	Three models of succession: facilitation, tolerance, inhibition	HIGH
F.E. Clements	1916	Monoclimax theory of succession; relay floristics	MEDIUM

Key Formulae — Part 1

Formula	Variables	Use
$dN/dt = rN$	N = population size, r = intrinsic rate of increase	Exponential growth rate
$dN/dt = rN(K–N)/K$	K = carrying capacity	Logistic growth rate
Max growth at N = K/2	—	Maximum growth rate in logistic model; also = MSY
R■ = Σ(lx × mx)	lx = survival to age x; mx = fecundity at age x	Net reproductive rate
$\lambda = e^r$	r = intrinsic rate of increase	Finite rate of increase per unit time
$N = M \times n / m$	M = marked released; n = total recaptured; m = recaptured marked	Lincoln-Petersen mark-recapture estimate
$H' = -\Sigma(p_i \times \ln p_i)$	pi = proportion of species i	Shannon-Wiener diversity index
$\beta = \gamma/\alpha$	γ = gamma; α = alpha diversity	Whittaker's beta diversity

Mnemonics — Part 1

Topic	Mnemonic / Memory Aid
Liebig vs Shelford	LIEBIG = LEAST (scarcest). SHELFORD = SPECTRUM (min–opt–max for each factor)

Topic	Mnemonic / Memory Aid
Population growth max rate	Max logistic growth = K/2. Also = MSY in fisheries. Never K, never 0.
r vs K selection	r = Rabbit (many, fast, small). K = Killer whale (few, slow, big, smart)
Survivorship curves	Type I = INVEST in few (humans). Type II = CONSTANT loss (birds). Type III = SCATTER many, hope some survive (fish)
Succession pioneer on rock	LICHENS on LIMESTONE. 'LL rule'. Crustose lichens are hardest-core pioneers.
Hydrarch sequence	P-S-F-R-M-Sh-F = Phytoplankton, Submerged, Floating, Reed, Marsh, Shrub, Forest
Psammosere pioneer	Marram grass = Ammophila (sounds like 'ammo' = tough as ammo, stabilises sand)
Species interactions signs	++ Mutualism, +0 Commensalism, –0 Amensalism, +– Predation/Parasitism, –– Competition
Gause experiment organism	PARAMECIUM proves competitive exclusion. P. aurelia wins over P. caudatum.
Alpha Beta Gamma	ABC: Alpha = local, Beta = Between, Gamma = Grand total

WHAT'S IN PART 2 — Environmental Biology (Topics 6–10)

Part 2 covers: Biodiversity (hotspots, HIPPO threats, endemic species in detail), Conservation Biology (in-situ/ex-situ, IUCN categories, protected areas, wildlife laws), Population Genetics (Hardy-Weinberg, genetic drift, founder effect, bottleneck, inbreeding depression), Evolution (natural selection types, speciation, coevolution), Applied Ecology (island biogeography, metapopulation, ecological footprint, ecosystem services valuation). Request Part 2 to complete the full Unit 3 coverage.